

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (CL)

May 2018

CL 306 Transportation Engineering II

Date: 01.05.2018, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

**Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

**SECTION - I**

- Q - 1 (a)** What is the significance of transportation engineering in nation's development? [05]  
**(b)** Enlist the classification of signals. [02]
- Q - 2 (a)** Draw a neat sketch of a cross-section of a double line broad gauge railway track in embankment. [04]  
**(b)** Explain Fish Plates with neat sketch. [05]  
**(c)** Write short note on Curves provided for designing of geometrics on railway track. [05]

**OR**

- (c)** Explain Gradient. [05]
- Q - 3 (a)** What is Transition Curve? Why Transition Curves are provided on Railway Tracks? [07]  
Enlist types of Transition Curves.
- (b)** Write Short Note on Turnout with neat sketch. [07]

**SECTION – II**

- Q - 4** Define the following terms: [07]  
(1) Taxiway (2) Terminal Building (3) Hanger (4) Pier (5) Jetty  
(6) Ferry (7) Transit Shed

- Q - 5 (a)** Enlist the types of harbours according to its classification. [04]

**OR**

- Q – 5 (a)** Enlist the design factors of harbour. Explain any one. [04]  
**(b)** Write Short Note on Dry Docks. [05]  
**(c)** Define aircraft wheel configuration with neat sketch. [05]
- Q-6 (a)** What are the points to be kept in mind during the selection of site for constructing an [04]

airport?

- (b)** Write a note on pattern of runways with neat sketch. **[04]**

**OR**

- (b)** Write a note on Wind Rose Diagram. **[04]**

- (c)** Write short note on navigational aids of airport. **[06]**

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